

Notes: DeMarzo & Duffie (Econometrica, 1999)

A Liquidity-Based Model of Security Design

Contents

1	Big picture	3
2	Model environment and key objects	3
2.1	Assets, issuer, and retention cost	4
2.2	Security design and contractible information	4
2.3	Private information at the sale stage	4
2.4	Issuer's liquidation problem for a fixed security	5
2.5	Ex-ante security-design problem	5
2.6	Timing	5
3	Market demand and reduced-form issuer profit (all equations + interpretation)	5
3.1	General properties of the public-offering profit function	5
3.2	Signalling equilibrium and endogenous illiquidity	6
3.3	Reduced-form profit from issuance	6
3.4	Why f_0 matters so much	7
3.5	Economic reading of the figures in Section 3	7
4	Properties of optimal security design	7
4.1	Marginal value of adding cash flows to a design	7
4.2	An optimal design must pay all available cash flow in some states	8
4.3	When pure equity is optimal	8
4.4	Only highly sensitive states should be left out of the security	8
4.5	If private information is contractible, the lemons problem can be neutralized	9
5	Standard debt	9
5.1	Additional restrictions in the debt section	9
5.2	Uniform worst case	9
5.3	Why standard debt is optimal	10
5.4	The optimal debt is risky, not riskless	10
5.5	Comparative statics for the face value of debt	10
5.6	Numerical example and the economic message of Figures 6 and 7	11
6	Pre-design information, short summary, and conclusion	11

6.1 Pre-design information 11

6.2 Short summary 11

6.3 Conclusion 12

1 Big picture

- **Question.** What security should an issuer design when it wants to raise cash by securitizing a fixed asset pool, but the issuer or underwriter will later possess private information that makes the security hard to sell at a fair price?
- **Economic setting.** The issuer owns assets with future cash flow X . It can design a security F backed by those assets and sell some fraction of the issue to outside investors. The issuer values retained future cash flows less than cash today because it has a private discount factor $\delta \in (0, 1)$.
- **Core friction.** After the design is chosen, the seller learns private information Z about the payoff of the security. Investors understand this and interpret a large sale quantity as bad news. Retention becomes a costly signal of quality, so market demand is downward sloping.
- **Main tradeoff.** A design that pays out more of the underlying cash flow raises more capital, but it is usually more sensitive to the seller's private information and therefore more illiquid. A safer design leaves more cash flow with the issuer, which is costly because retained cash flows are discounted by δ .
- **Main conceptual contribution.** The paper endogenizes liquidity through a signalling model. Illiquidity is not imposed from outside; it arises because better-informed sellers must retain part of the issue to convince buyers that the security is not a lemon.
- **Key reduced-form result.** For any security design, the issuer's public-offering profit can be summarized by a function $\mathcal{H}(f, f_0)$, where $f = E(F | Z)$ is the seller's private valuation and $f_0 = \min_z E_z(F)$ is the worst-case private valuation. This profit function is decreasing, convex, and homogeneous of degree one.
- **Main design results.** An optimal security must exhaust the full asset cash flow in some states; if informational sensitivity is sufficiently low, pure equity is optimal; and if payoffs must be monotone in asset cash flow and a *uniform worst case* exists, then **standard debt is optimal**.
- **Debt result.** The optimal debt is generally *risky*, not riskless. The issuer chooses a face value that trades off lemons costs on risky debt against the retention cost of leaving too much cash flow unsecuritized.
- **Comparative statics.** Higher asset volatility or more severe asymmetric information lowers both securitization profits and the optimal face value of debt. Higher retention costs push the issuer toward riskier debt or, in the limit, pure equity.
- **Core takeaway.** Debt is attractive here because it is the monotone security with the *least informational sensitivity*. The paper provides a liquidity-based foundation for debt design that is distinct from taxes, bankruptcy costs, or traditional capital-structure stories.

2 Model environment and key objects

2.1 Assets, issuer, and retention cost

- The issuer owns assets generating a bounded nonnegative random cash flow X .
- All agents are risk neutral and the market interest rate is normalized to zero.
- The issuer values one dollar of retained future payoff at only $\delta \in (0, 1)$ dollars today.

Interpretation. The issuer wants cash now because of capital requirements, funding pressure, or profitable outside investment opportunities. The parameter δ is the shadow value of keeping cash flows on balance sheet rather than selling them.

2.2 Security design and contractible information

- A security payoff is a limited-liability claim

$$F = \varphi(S), \quad 0 \leq F \leq X,$$

where S is publicly observable contractible information.

- If the security is conditioned only on realized asset cash flow, then $S = X$ and the design is simply $F = \varphi(X)$.
- More generally, S can include publicly verifiable indices, prepayment behavior, sales, market prices, or other contractible variables.

Interpretation. The paper allows a broad class of state-contingent securities. Standard debt and equity are just special cases inside a much larger design space.

2.3 Private information at the sale stage

- After the design is chosen but before the security is sold, the issuer learns private information Z .
- This information changes the issuer's conditional valuation of the security:

$$f \equiv E(F \mid Z).$$

- The support of the distribution of f is written $[f_0, f_1]$, where

$$f_0 \equiv \min_z E_z(F).$$

Interpretation. The whole lemons problem is driven by variation in f . If f were constant, investors would not worry that the issuer is selling only when the security is weak.

2.4 Issuer's liquidation problem for a fixed security

- Let $P_F(q)$ be the market price per unit of the security when the issuer sells a fraction $q \in [0, 1]$ of the issue.
- If the issuer sells fraction q , its total value is

$$\delta E(X - F | Z) + \delta(1 - q)E(F | Z) + qP_F(q) = \delta E(X | Z) + q[P_F(q) - \delta E(F | Z)].$$

- Given private valuation $f = E(F | Z)$, the sale problem is

$$\mathcal{H}_F(f) = \sup_{q \in [0, 1]} q[P_F(q) - \delta f]. \quad (1)$$

Interpretation. This is a monopoly problem. The issuer's marginal cost of selling one more unit is the private value of retaining it, δf . The shape of $P_F(q)$ therefore determines how severe the liquidity problem is.

2.5 Ex-ante security-design problem

Once the liquidation stage is summarized by $\mathcal{H}_F$, the issuer chooses the design to maximize expected securitization profits:

$$V(F) = E[\mathcal{H}_F(E(F | Z))], \quad \sup_F V(F). \quad (2)$$

Interpretation. The security design matters only because it changes two objects at once: how much cash flow is packaged into the security, and how sensitive that packaged cash flow is to the seller's private information.

2.6 Timing

The paper's basic timing is:

asset pool $X \rightarrow$ security F chosen $\rightarrow$ private information Z realized $\rightarrow$ quantity sold $q \rightarrow$ price
 $P_F(q) \rightarrow$ final payoffs.

Interpretation. The design comes *before* the issuer learns the sale-stage signal. This is why the issuer chooses a design that is robust to future lemons problems.

3 Market demand and reduced-form issuer profit (all equations + interpretation)

3.1 General properties of the public-offering profit function

For any given demand schedule P_F , Proposition 1 shows:

- $\mathcal{H}_F(f)$ is **decreasing** in f ;
- $\mathcal{H}_F(f)$ is **convex** in f ;
- any optimal quantity schedule $Q(f)$ is weakly **decreasing** in f .

Interpretation.

- When the issuer privately knows the security is valuable, it sells less.
- Convexity is crucial: the cost of information sensitivity rises disproportionately as the design becomes more exposed to private information.
- This convexity later explains why safer claims such as debt can dominate more informationally sensitive claims.

3.2 Signalling equilibrium and endogenous illiquidity

Investors observe only the fraction sold, not the issuer's private signal. A separating Bayes–Nash equilibrium therefore satisfies:

$$Q(f) \in \arg \max_{q \in [0,1]} q[P(q) - \delta f], \quad (3)$$

$$P(Q(f)) = E[f \mid Q(f)], \quad (4)$$

$$P(Q(f)) = f \quad \text{in a separating equilibrium.} \quad (5)$$

For the continuous separating equilibrium used in the paper,

$$Q^*(f) = \left(\frac{f_0}{f}\right)^{\frac{1}{1-\delta}}, \quad P^*(q) = f_0 q^{-(1-\delta)}. \quad (6)$$

Interpretation.

- Better issuer information (higher f) leads to more retention and smaller quantity sold.
- Market price falls with quantity sold because investors infer that larger sales come from weaker issuer valuations.
- Retention is a costly signal because unsold claims are worth only δf to the issuer.

3.3 Reduced-form profit from issuance

Substituting the separating equilibrium into the issuer's payoff gives a reduced-form public-offering profit function

$$\mathcal{H}(f, f_0) = U(f, Q^*(f), f) = (1 - \delta)f \left(\frac{f_0}{f}\right)^{\frac{1}{1-\delta}} \quad (7)$$

for the continuous case. The paper emphasizes that the rest of the analysis uses only the *properties* of $\mathcal{H}$, not this exact formula.

Proposition 3 shows:

1. $\mathcal{H}$ is homogeneous of degree one;
2. $\mathcal{H}(f_0, f_0) = (1 - \delta)f_0$ and $\mathcal{H}(f, 0) = 0$;
3. for $f > f_0 > 0$, $\mathcal{H}(f, f_0)$ is strictly decreasing and strictly convex in f ;
4. the derivative satisfies $-\delta < \mathcal{H}_1(f, f_0) < 0$.

Interpretation.

- **Homogeneity** means that scaling up all payoffs scales up optimal profits proportionally.
- **Decreasingness** means that states in which the issuer privately values the claim more are precisely the states in which it has to retain more, so sale profits are lower.
- **Convexity** means informational sensitivity is especially costly in bad liquidity environments.

3.4 Why f_0 matters so much

A central simplification is that the signalling equilibrium depends on the design only through f_0 , the lowest possible seller valuation. **Interpretation.** Once the market understands the worst possible type, every other type chooses quantity relative to that benchmark. This lets the paper compare complicated security designs through a relatively low-dimensional object.

3.5 Economic reading of the figures in Section 3

- **Figure 2.** The issuer's sale profit is the upper envelope of affine functions in f , so the envelope is decreasing and convex.
- **Figure 3.** At $f = f_0$ the issuer sells the whole issue and gains the full holding-cost benefit $(1 - \delta)f_0$. As f/f_0 rises, the issuer must retain more of the issue, and profits fall sharply.

Economic message. Liquidity costs are endogenous and nonlinear. A design that is only slightly more informationally sensitive can be much harder to sell.

4 Properties of optimal security design

4.1 Marginal value of adding cash flows to a design

Take a candidate design F and an incremental nonnegative cash-flow component G . Let

$$A = \{X - F > 0\}$$

be the states in which the security does *not* already exhaust available cash flow. The paper defines the marginal private valuation

$$y(F, G) = E(G1_A \mid Z)$$

and the marginal worst-case valuation

$$y_0(F, G) = \min_{z \in S^*} E_z(G1_A), \quad S^* = \{z : E_z(F) = f_0\}.$$

For $f_0 > 0$ and $y_0(F, G) > 0$, equation (6) writes the directional derivative of issuer value as

$$VV(F; G) = y_0(F, G)E \left[\mathcal{H}_1\left(\frac{f}{f_0}, 1\right) \left(\frac{y(F, G)}{y_0(F, G)} - \frac{f}{f_0} \right) + \mathcal{H}\left(\frac{f}{f_0}, 1\right) \right]. \quad (8)$$

Interpretation. The key comparison is between the design's existing information sensitivity f/f_0 and the *marginal* information sensitivity y/y_0 of the added cash flow. If the added cash flow is relatively insensitive to private information, adding it is valuable.

4.2 An optimal design must pay all available cash flow in some states

Proposition 5 states that if F is optimal, then the event

$$\{F = X\}$$

has strictly positive probability. **Interpretation.** It cannot be optimal to leave every state only partially securitized. If there were always slack, the issuer could add a small risk-free increment and raise more cash with almost no extra lemons problem.

4.3 When pure equity is optimal

Define the informational sensitivity of S to Z as the maximal distortion in probabilities induced by conditioning on different seller information states. Let r be the excess return on cash relative to retained assets, defined by

$$\delta = \frac{1}{1 + r}.$$

Proposition 6 shows:

If the informational sensitivity of S to Z is strictly less than r , then the unique optimal design is **pure equity**, $F = X$.

Interpretation. When private information is mild, the lemons premium from issuing all cash flows is small relative to the issuer's benefit from raising cash immediately. In that case the issuer should securitize the whole asset pool.

4.4 Only highly sensitive states should be left out of the security

Proposition 7 strengthens the last point. If there is a measurable subset B of states on which $F < X$, then that set must itself be highly informationally sensitive. **Interpretation.** The issuer

only wants to hold back cash flows that are especially revealing of its private information. Safe or weakly informative cash flows should be put into the security.

4.5 If private information is contractible, the lemons problem can be neutralized

Suppose the seller's private information is itself contractible, meaning Z is measurable with respect to S . Proposition 9 shows that there exists an optimal security with

$$f = f_0 = \min_z E_z(X),$$

so the seller's private valuation is constant across information states and the issuer sells the whole issue. **Interpretation.** If the seller can write the contract directly on the information that would otherwise create adverse selection, then the security can be designed as a warranty against lemons. The optimal contract need not be riskless, but it can be designed so that the seller's information no longer moves its value.

5 Standard debt

5.1 Additional restrictions in the debt section

To obtain a sharp debt result, the paper adds two restrictions:

- the contractible signal is the asset cash flow itself, so $S = X$;
- attention is restricted to monotone securities $F = \varphi(X)$ with φ nondecreasing.

A standard debt contract with face value d is

$$F = \min(X, d).$$

If d is above every possible value of X , this collapses to pure equity $F = X$. **Interpretation.** In many securitization settings, actual cash flows are the natural contractible signal, and monotone payoffs are the economically relevant class.

5.2 Uniform worst case

The crucial statistical assumption is the existence of a *uniform worst case* z_0 . Informally, conditional on any interval of X , the distribution under any other seller type first-order stochastically dominates the distribution under z_0 . **Interpretation.** The worst seller-information state is uniformly bad across the whole support. This assumption is weaker than a full monotone likelihood ratio property and is enough to rank designs by informational sensitivity.

5.3 Why standard debt is optimal

Proposition 10: **If there is a uniform worst case, then a standard debt contract is optimal.**

The proof compares any monotone security $G = \varphi(X)$ with a debt contract $F = \min(X, d)$ chosen to have the same worst-case value $f_0 = g_0$. Debt then weakly reduces seller-information sensitivity at every type while preserving the worst-case benchmark that governs the equilibrium liquidity schedule. **Interpretation.**

- Debt carves out the safest part of the cash-flow distribution.
- Because it truncates upside states, debt depends less on the issuer's private information than equity-like claims do.
- Among monotone claims with the same worst-case payoff, debt is therefore the easiest to sell.

5.4 The optimal debt is risky, not riskless

Even when debt is optimal, the optimal face value need not be low enough to make the debt riskless. The issuer chooses d to solve

$$\max_d V(\min(X, d)).$$

Interpretation. Raising d securitizes more cash flows and raises more capital, but also makes debt more sensitive to private information and therefore more illiquid. The optimum is typically an interior, risky debt level.

5.5 Comparative statics for the face value of debt

Proposition 11 gives useful comparative statics.

- If cash flows are scaled multiplicatively, $X(b) = bY$, then the optimal face value scales proportionally: $D(b) = bD(1)$.
- If safe cash is added additively, $X(b) = b + Y$, then the optimal face value rises by less than one-for-one:

$$D(b) - D(0) \leq b,$$

with strict inequality under an interior solution.

Interpretation. Safer asset pools support debt, but the issuer does not mechanically translate every extra safe dollar into additional promised debt. The exact face value still balances retention costs and lemons costs.

5.6 Numerical example and the economic message of Figures 6 and 7

The paper studies a parameterized example with

$$X = (1 + Z)Y,$$

where Y is lognormal and Z is uniformly distributed on $[-m, m]$.

- **Figure 6.** For fixed volatility, larger information parameter m lowers securitization profit at every debt level and lowers the optimal face value of debt.
- **Figure 7.** For fixed information asymmetry, higher asset volatility lowers securitization profits and lowers the optimal face value of debt.

Economic message. Riskier firms may issue less debt not only because of bankruptcy costs, but also because risky debt is more exposed to adverse-selection discounts when sellers are better informed.

6 Pre-design information, short summary, and conclusion

6.1 Pre-design information

The paper extends the model to the case in which the issuer learns some information Z_a *before* choosing the design and learns further information Z_b only later, before sale. Under two conditions,

1. there exists a sufficient statistic $h(Z)$ such that S and Z are independent conditional on $h(Z)$;
2. the support of the conditional distribution of $h(Z)$ given Z_a does not depend on Z_a ,

the same design logic goes through after conditioning on Z_a . **Interpretation.** Early information changes which debt face value is optimal, but it does not destroy the structure of the problem. If early information is especially favorable, the issuer may choose a lower face value because it anticipates a more severe lemons premium at the sale stage.

6.2 Short summary

- The issuer designs a security before learning sale-stage private information.
- Because investors infer quality from the quantity sold, the seller faces a downward-sloping demand curve.
- The resulting public-offering profit is decreasing and convex in the seller's private valuation.
- Optimal designs include all low-information-sensitivity cash flows.

- If information sensitivity is small enough, pure equity is optimal.
- If payoffs must be monotone in X and a uniform worst case exists, standard debt is optimal.
- The optimal debt is generally risky, and its face value falls as information asymmetry or asset risk rises.

6.3 Conclusion

Core conclusion. The paper gives a clean theory of security design in which liquidity is shaped by asymmetric information at issuance. The design problem is not just about carving up cash flows; it is about choosing a claim that can be sold with minimal signalling distortion.

Economic intuition. Safer claims such as debt are valuable because they mute the mapping from private information to security value. That weakens the lemons problem and reduces the amount of costly retention needed to convince investors.

Broader implication. The strict convexity of the issuance-profit function also suggests a rationale for tranching and splitting claims: heterogeneous pieces can sometimes be sold more efficiently than one undifferentiated security. The paper therefore serves as a foundation for later work on capital structure, securitization, and tranche design.